OIL PAN (Continued)

INSTALLATION

(1) Install the oil pan gasket to the block.

(2) Apply a 3MM (1/8 inch) bead of Mopar® Engine RTV at the oil pump to engine block parting line (Fig. 82).

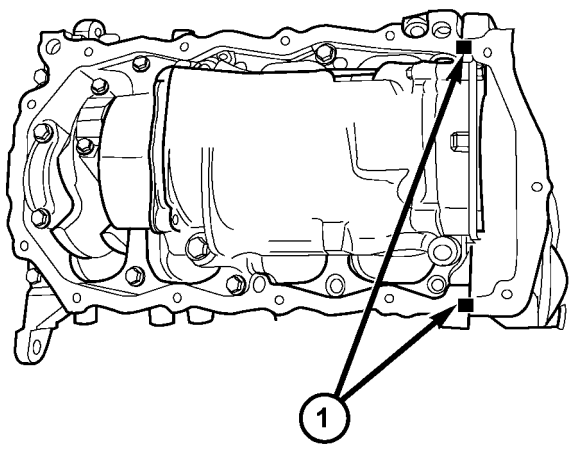

Fig. 82 OIL PAN GASKET INSTALLATION

1 - SEALER LOCATION

(3) Install pan and tighten the screws to 12 N·m (105 in. lbs.).

(4) Lower engine, and remove Special Tool 8534.

(5) Tighten engine mount thru bolts.

(6) Raise the front axle into position, and reinstall front axle mounting bolts. If equipped.

(7) Reconnect exhaust pipe to manifold.

(8) Install structural collar, if equipped.

(9) Lower vehicle.

(10) Fill engine crankcase with proper oil to correct level.

(11) Reinstall air cleaner assembly.

OIL PRESSURE SENSOR/ SWITCH

DESCRIPTION

The 3–wire, electrical/mechanical engine oil pressure sensor (sending unit) is located in an engine oil pressure gallery.

OPERATION

The oil pressure sensor uses three circuits. They are:

- A 5 volt power supply from the Powertrain Control Module (PCM)
- A sensor ground through the PCM's sensor return
- A signal to the PCM relating to engine oil pressure

The oil pressure sensor has a 3 wire electrical function very much like the Manifold Absolute Pressure (MAP) sensor. Meaning different pressures relate to different output voltages.

A 5 volt supply is sent to the sensor from the PCM to power up the sensor. The sensor returns a voltage signal back to the PCM relating to engine oil pressure. This signal is then transferred (bussed) to the instrument panel on either a CCD or PCI bus circuit (depending on vehicle line) to operate the oil pressure gauge and the check gauges lamp. Ground for the sensor is provided by the PCM through a low-noise sensor return.

OIL PUMP

REMOVAL

(1) Disconnect negative cable from battery.

(2) Remove timing belt. (Refer to 9 - ENGINE/ VALVE TIMING/TIMING BELT/CHAIN AND SPROCKETS - REMOVAL)

(3) Remove timing belt rear cover. (Refer to 9 - ENGINE/VALVE TIMING/TIMING BELT / CHAIN COVER(S) - REMOVAL)

(4) Remove oil pan. (Refer to 9 - ENGINE/LUBRICATION/OIL PAN - REMOVAL)

(5) Remove crankshaft sprocket using Special Tools 6793 and C-4685-C2 (Fig. 83).

(6) Remove crankshaft key (Fig. 84).

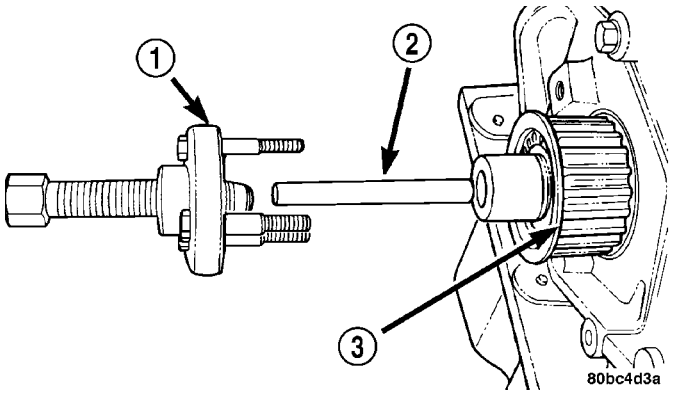

Fig. 83 Crankshaft Sprocket - Removal

1 - SPECIAL TOOL 6793
2 - SPECIAL TOOL C-4685–C2
3 - CRANKSHAFT SPROCKET

(7) Remove oil pick-up tube.

(8) Remove oil pump (Fig. 85) and front crankshaft seal.